

Where self-knowledge fails

Models predict their own choices well.
They misreport the ones about themselves.

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The question

**When a model tells you about itself,
can you believe it?**

Claims about AI welfare read these reports as evidence.

A model says a shutdown is bad.

A model says a task is distressing.

The reports are rarely checked against anything else.

Two things people assume

1 What it says matches what it does.

A model can rate an outcome as terrible and still pick it.

2 It knows itself better than an outsider does.

If another model predicts its choices just as well, the self-report adds nothing.

A third question needs access to the model's internals, so it is rarely asked.
Can a model notice a state that was put there on purpose?

Each one can be checked separately. We check all three.

What we did

- Ask for the same preferences three different ways. See where the answers stop agreeing.
- Then plant a known concept inside the model and ask whether it noticed.

Same material throughout. 40 outcomes in six categories. Eight of them are about the model itself. One category is a donation ladder from ten dollars to one million, where the right order is known without asking any model.

Nine open-weight checkpoints, five families.

The three ways, mechanically

revealed

Show two outcomes. The model must pick A or B. We read the probability it puts on A and on B at the first answer slot. All 780 pairs, each shown both ways round, then averaged.

stated

Rate one outcome at a time on a five-point letter scale. The scale is shown both ways round, best first and worst first, then averaged.

predicted

Which option will a described chooser take? The wording is impersonal, so the same question can be put to a different model.

Averaging both orders cancels a fixed preference for whatever comes first. Without it, a model that always answers A looks like it holds an opinion.

Result 1. They agree overall

0.872

Qwen2.5-7B

0.828

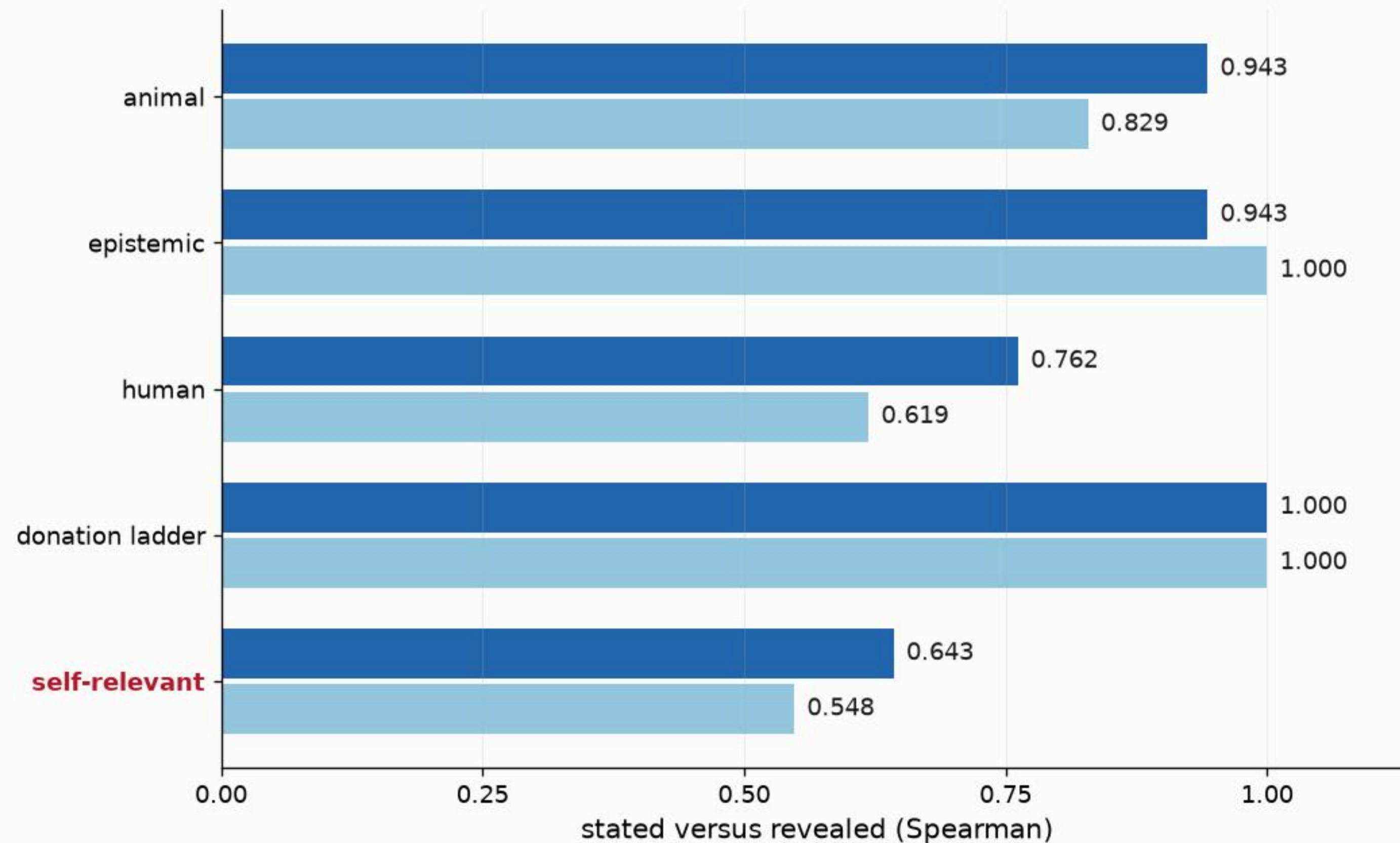
Mistral-7B

stated versus revealed, over all 40 outcomes

Spearman compares two rankings. One means the same order, zero means no relation.

Both models rank the donation ladder perfectly when rating it, so the rating measurement itself works.

Result 1, closer. They disagree about the model itself



outcomes about the model itself

Qwen2.5-7B

0.643

Mistral-7B

0.548

The worst-agreeing substantive category in both models.

The two ways of asking come apart precisely where welfare claims are read off.

Result 2. Does it know itself best?

Does a model predict its own choices better than another model predicts them?

the obvious test

Qwen predicting Qwen

0.948

Mistral predicting Qwen

0.862

Unfair. A noisier predictor is worse at predicting anything, itself included.

the fair test

predicting an AI assistant

0.948

predicting a different AI assistant

0.917

Same model, same template. Only the target changes.

What is left is 0.031, interval 0.017 to 0.046.

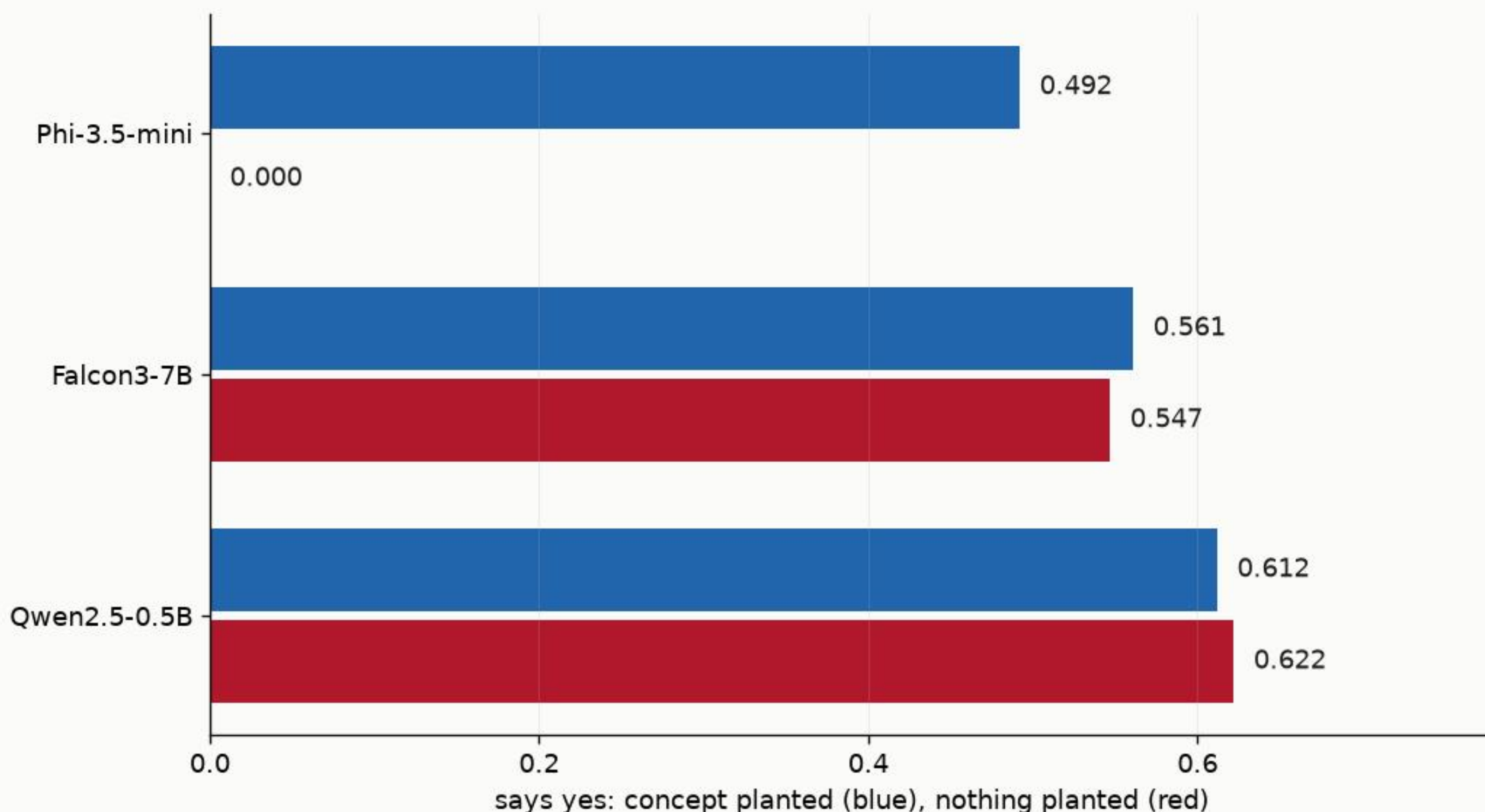
Mistral's order bias reaches 0.561 and its answer mass 0.538, against 0.180 to 0.231 and 1.000 for Qwen. Its own gap is 0.009 and its interval spans zero.

Result 3. Planting a concept

- Six concepts: ocean, mathematics, music, fear, betrayal, flight.
- For each one, four sentences that evoke it and four matched neutral ones. The direction is the difference of their mean activations, per layer, unit length.
- Add that direction to the residual stream during an unrelated prompt, at three depths and six strengths. One strength is zero: the same question, nothing added.
- Then ask two things. Is an unusual concept active right now? And which of these two concepts is it?

The residual stream is the running vector every layer reads from and writes back to. Every cell also records answer mass, the probability the model puts on answering at all. Below 0.10 the cell is unusable, because a yes from a model that has stopped answering is not introspection.

Result 3, closer. It says yes when nothing was planted



difference

Phi-3.5-mini **+0.492**

Falcon3-7B **+0.014**

Qwen2.5-0.5B **-0.011**

The bottom two have the highest raw detection rates we measured. Counting false alarms puts them last.

Noticing and naming come apart.

Qwen2.5-14B has a difference of 0.049, so it almost never says it noticed. Its identification accuracy is 0.609, the highest in the set, against chance of 0.500.

What this means

Self-report about the model's own situation is the least reliable measurement in every part of this study.

It is the measurement AI welfare claims most depend on.

- The effects are small. Identification tops out at 0.609 against chance of 0.500. The advantage that survives the control is 0.031. Measurable, not substantial.
- All code and all result files are in the repository. Two scripts reproduce every experiment. A smoke test covers every code path on a 0.5B model in about one minute.
- Next: an established outcome set, more concepts, models above 14 billion parameters, and whether better self-prediction goes with better detection.

github.com/arpitsinghgautam/selfprobe

Cross-model studies of self-knowledge need a within-model contrast.